

第二次习题课参考解答(复合函数链式法则、高阶偏导数、方向导数)

1. 解答下列各题:

(1) 设 f 可微, 且 $z = x^3 f\left(xy, \frac{y}{x}\right)$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

$$\begin{aligned}\text{解: } dz &= f \cdot 3x^2 dx + x^3 df = 3x^2 f dx + x^3 \left[f'_1 d(xy) + f'_2 d\left(\frac{y}{x}\right) \right] \\ &= 3x^2 f dx + x^3 \left[f'_1 (x dy + y dx) + f'_2 \frac{xdy - ydx}{x^2} \right] \\ &= (3x^2 f + x^3 y f'_1 - x y f'_2) dx + (x^4 f'_1 + x^2 f'_2) dy\end{aligned}$$

由一阶微分的形式不变性,

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = (3x^2 f + x^3 y f'_1 - x y f'_2) dx + (x^4 f'_1 + x^2 f'_2) dy$$

$$\text{故 } \frac{\partial z}{\partial x} = (3x^2 f + x^3 y f'_1 - x y f'_2), \quad \frac{\partial z}{\partial y} = (x^4 f'_1 + x^2 f'_2).$$

其中符号 f'_1 , f'_2 分别表示函数 $f(x, y)$ 对第一个中间变量和第二个中间变量求偏导。

(2) 设 $z = f(x^2 y, \frac{y}{x})$, 其中 $f \in C^2$, 求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial^2 z}{\partial y \partial x}$.

$$\begin{aligned}\text{解: } \frac{\partial z}{\partial x} &= 2xy f'_1 - \frac{y}{x^2} f'_2; \\ \frac{\partial^2 z}{\partial y \partial x} &= 2xy (f''_{11} x^2 + \frac{1}{x} f''_{12}) + 2x f'_1 - \frac{y}{x^2} (x^2 f''_{21} + \frac{1}{x} f''_{22}) - \frac{1}{x^2} f'_2 \\ &= 2x^3 y f''_{11} + y f''_{12} - \frac{y}{x^3} f''_{22} + 2x f'_1 - \frac{1}{x^2} f'_2.\end{aligned}$$

(3) 设 $g(x) = f(x, \phi(x^2, x^2))$, 其中函数 f 和 ϕ 的二阶偏导数连续, 求 $g''(x)$.

解: 由 $g(x) = f(x, \phi(x^2, x^2))$ 两边对 x 求导, 得

$$g'(x) = f'_x(x, \phi(x^2, x^2)) + 2 f'_\phi(x, \phi(x^2, x^2)) (\phi'_1(x^2, x^2) + \phi'_2(x^2, x^2)) x,$$

两边再对 x 求导, 得

$$g''(x) = f''_{xx} + 4 f''_{x\phi} (\phi'_1 + \phi'_2) x + 4 f''_{\phi\phi} (\phi'_1 + \phi'_2)^2 x^2 + 4 f'_\phi (\phi''_{11} + 2\phi''_{12} + \phi''_{22}) x^2 + 2 f'_\phi (\phi'_1 + \phi'_2),$$

其中符号 ϕ'_1 , ϕ'_2 分别表示 ϕ 对其第一个中间变量和第二个中间变量求偏导。

(4) 设 $u(x, y) \in C^2$, 又 $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$, $u(x, 2x) = x$, $u'_x(x, 2x) = x^2$, 求 $u''_{xx}(x, 2x)$,

$u''_{xy}(x, 2x)$, $u''_{yy}(x, 2x)$.

解: 因为 $\frac{\partial u}{\partial x}(x, 2x) = x^2$, 两边对 x 求导, 得

$$\frac{\partial^2 u}{\partial x^2}(x, 2x) + 2 \frac{\partial^2 u}{\partial y \partial x}(x, 2x) = 2x. \quad (1)$$

由 $u(x, 2x) = x$, 两边对 x 求导, 得 $\frac{\partial u}{\partial x}(x, 2x) + 2 \frac{\partial u}{\partial y}(x, 2x) = 1$,

所以, $\frac{\partial u}{\partial y}(x, 2x) = \frac{1-x^2}{2}$. 此式两边再对 x 求导, 得

$$\frac{\partial^2 u}{\partial x \partial y}(x, 2x) + 2 \frac{\partial^2 u}{\partial y^2}(x, 2x) = -x. \quad (2)$$

由已知,

$$\frac{\partial^2 u}{\partial x^2}(x, 2x) - \frac{\partial^2 u}{\partial y^2}(x, 2x) = 0, \quad (3)$$

因为 $u(x, y) \in C^2$, 因此 $\frac{\partial^2 u}{\partial x \partial y}(x, 2x) = \frac{\partial^2 u}{\partial y \partial x}(x, 2x)$.

联立 (1), (2), (3) 式, 解得

$$\frac{\partial^2 u}{\partial x^2}(x, 2x) = \frac{\partial^2 u}{\partial y^2}(x, 2x) = -\frac{4}{3}x, \quad \frac{\partial^2 u}{\partial x \partial y}(x, 2x) = \frac{5}{3}x.$$

(5) 已知 $z = \left(\frac{y}{x}\right)^{\frac{x}{y}}$, 求 $\frac{\partial z}{\partial x}\bigg|_{(1,2)}$.

解: 令 $u = \frac{y}{x}$, $v = \frac{x}{y}$, 则 $z = u^v$. 所以

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = v u^{v-1} \left(-\frac{y}{x^2}\right) + \frac{1}{y} u^v \ln u.$$

因为 $u(1, 2) = 2$, $v(1, 2) = \frac{1}{2}$, 因此 $\frac{\partial z}{\partial x}\bigg|_{(1,2)} = \frac{\ln 2 - 1}{\sqrt{2}}$.

(6) 设函数 f 二阶可导, 函数 g 一阶可导. 令

$$z(x, y) = f(x+y) + f(x-y) + \int_{x-y}^{x+y} g(t) dt. \text{ 求 } \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}.$$

解: 由复合函数求导法则及变限积分求导, 可得

$$\frac{\partial z}{\partial x} = f'(x+y) + f'(x-y) + g(x+y) - g(x-y),$$

$$\frac{\partial z}{\partial y} = f'(x+y) - f'(x-y) + g(x+y) + g(x-y),$$

$$\text{所以 } \frac{\partial^2 z}{\partial x^2} = f''(x+y) + f''(x-y) + g'(x+y) - g'(x-y),$$

$$\frac{\partial^2 z}{\partial y^2} = f''(x+y) + f''(x-y) + g'(x+y) + g'(x-y),$$

$$\frac{\partial^2 z}{\partial x \partial y} = f''(x+y) - f''(x-y) + g'(x+y) + g'(x-y).$$

$$(7) \text{ 设 } f(x, y) = \begin{cases} e^{-\frac{1}{x^2+y^2}}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0. \end{cases} \text{ 求 } f''_{xx}(0, 0) \text{ 与 } f''_{xy}(0, 0).$$

$$\text{解: } f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0,$$

$$\text{对 } \forall (x, y) \neq (0, 0), \quad f'_x(x, y) = \frac{2x}{(x^2 + y^2)^2} e^{-\frac{1}{x^2+y^2}},$$

$$\text{因此 } f''_{xx}(0, 0) = \lim_{x \rightarrow 0} \frac{f'_x(x, 0) - f'_x(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{2e^{-\frac{1}{x^2}}}{x^4} = 0;$$

$$f''_{xy}(0, 0) = \lim_{y \rightarrow 0} \frac{f'_x(0, y) - f'_x(0, 0)}{y} = 0.$$

$$(8) \text{ 设 } f(x, y) = e^x \sin y, \text{ 求 } f^{(n+m)}_{x^n y^m}(0, 0).$$

解: 因为 $f^{(n+m)}_{x^n y^m}(x, y) = e^x \sin(y + \frac{m}{2}\pi)$, 因此

$$f^{(n+m)}_{x^n y^m}(0, 0) = \sin \frac{m}{2}\pi = \begin{cases} 0, & m = 2k, \\ (-1)^{k-1}, & m = 2k-1, \end{cases} \text{ 其中 } k \in \mathbb{N}^+.$$

2. 设 $z = z(x, y)$ 二阶连续可微, 并且满足方程 $A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} = 0$, 其中

A, B, C 都是非零常数。若令 $\begin{cases} u = x + \alpha y \\ v = x + \beta y \end{cases}$, 试确定 α, β 为何值时原方程可转化为

$$\frac{\partial^2 z}{\partial u \partial v} = 0.$$

解: 因为 $z = z(x, y)$ 二阶连续可微, 因此二阶混合偏导与求偏导顺序无关。将 z 看成以 u, v 为中间变量, x, y 为自变量的二元复合函数, 利用链式法则得

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = \alpha \frac{\partial z}{\partial u} + \beta \frac{\partial z}{\partial v},$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\alpha \frac{\partial z}{\partial u} + \beta \frac{\partial z}{\partial v} \right) = \alpha^2 \frac{\partial^2 z}{\partial u^2} + 2\alpha\beta \frac{\partial^2 z}{\partial u \partial v} + \beta^2 \frac{\partial^2 z}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\alpha \frac{\partial z}{\partial u} + \beta \frac{\partial z}{\partial v} \right) = \alpha \frac{\partial^2 z}{\partial u^2} + (\alpha + \beta) \frac{\partial^2 z}{\partial u \partial v} + \beta \frac{\partial^2 z}{\partial v^2},$$

由 $0 = A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2}$ 得到

$$(A + 2B\alpha + C\alpha^2) \frac{\partial^2 z}{\partial u^2} + 2(A + B(\alpha + \beta) + C\alpha\beta) \frac{\partial^2 z}{\partial u \partial v} + (A + 2B\beta + C\beta^2) \frac{\partial^2 z}{\partial v^2} = 0 \quad \dots (*)$$

故只要选取 α, β 使得

$$\begin{cases} A + 2B\alpha + C\alpha^2 = 0 \\ A + 2B\beta + C\beta^2 = 0, \end{cases}$$

即得 $\frac{\partial^2 z}{\partial u \partial v} = 0$. 这样问题转化为方程 $A + 2Bt + Ct^2 = 0$ 有两不同实根, 即要求

$B^2 - AC > 0$. 取 $\alpha = \frac{-B + \sqrt{B^2 - AC}}{C}, \beta = \frac{-B - \sqrt{B^2 - AC}}{C}$. 将其代入方程(*), 可知

$\frac{\partial^2 z}{\partial u \partial v}$ 的系数不为零, 从而 $\frac{\partial^2 z}{\partial u \partial v} = 0$.

3. 设 $z(x, y)$ 是定义在矩形区域 $D = \{(x, y) | 0 \leq x \leq a, 0 \leq y \leq b\}$ 上的可微函数。证明:

$$(1) \quad z(x, y) = f(y) \Leftrightarrow \forall (x, y) \in D, \frac{\partial z}{\partial x} \equiv 0;$$

$$(2) \quad z(x, y) = f(y) + g(x) \Leftrightarrow \forall (x, y) \in D, \frac{\partial^2 z}{\partial x \partial y} \equiv 0.$$

证明: (1) “ $\Rightarrow$ ” 显然.

“ $\Leftarrow$ ” 任取 $x_0 \in [0, a]$. 任意固定 $y \in [0, b]$, 将 $z(x, y)$ 看成关于 x 的一元函数在

以 x 与 x_0 为端点的区间上应用微分中值定理, 则存在 ξ 使得

$$z(x, y) - z(x_0, y) = \frac{\partial z}{\partial x}(\xi, y)(x - x_0) = 0,$$

这样 $z(x, y) = z(x_0, y)$, 故 $z(x, y) = f(y)$ 与 x 无关.

(2) “ $\Rightarrow$ ” 显然。

“ $\Leftarrow$ ” 因为 $\frac{\partial^2 z}{\partial x \partial y} \equiv 0$, $\frac{\partial z}{\partial y} = h(y)$ 与 x 无关. 故

$$z(x, y) = \int h(y) dy + g(x) = f(y) + g(x).$$

4. 证明: 函数 $f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ 在 $(0, 0)$ 点不连续, 但在该点存在任意

阶偏导数 $\frac{\partial^n f}{\partial x^n}(0, 0)$ 和 $\frac{\partial^n f}{\partial y^n}(0, 0)$.

证明: 令 $y = kx$, 则 $\lim_{(x, y) \rightarrow (0, 0)} f(x, kx) = \frac{k}{1+k^2}$ 依赖于 k 的值, 因此 $f(x, y)$ 在 $(0, 0)$ 的极限

不存在, 从而在该点不连续. 由于 $f(x, 0) = 0$, $f(0, 0) = 0$, 因此

$$f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0.$$

当 $x \neq 0$ 时,

$$f'_x(x, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, 0) - f(x, 0)}{\Delta x} = 0.$$

假设对任意的 x , 有 $\frac{\partial^{n-1} f}{\partial x^{n-1}}(x, 0) = 0$, 则

$$\frac{\partial^n f}{\partial x^n}(0,0) = \lim_{x \rightarrow 0} \frac{\frac{\partial^{n-1} f}{\partial x^{n-1}}(x,0) - \frac{\partial^{n-1} f}{\partial x^{n-1}}(0,0)}{x} = 0.$$

由于 $f(x,y)$ 的表达式中 x, y 的地位对称, 同理可得 $\frac{\partial^n f}{\partial y^n}(0,0) = 0$. 证毕

5. 设 $u(x,y)$ 有二阶偏导数, 无零点. 证明: $u(x,y)$ 满足方程 $u \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y}$ 当且仅当

$$u(x,y) = f(x)g(y).$$

证明: 充分性显然, 下证必要性. 假设 $u \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y}$. 由于 $u(x,y)$ 没有零点, 因此有

$$\frac{u \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial u}{\partial x} \frac{\partial u}{\partial y}}{u^2} = 0,$$

即 $\frac{\partial \left(\frac{1}{u} \frac{\partial u}{\partial y} \right)}{\partial x} = 0$. 从而 $\frac{1}{u} \frac{\partial u}{\partial y} = r(y)$, 故 $\frac{\partial(\ln u)}{\partial y} = r(y)$; 两端关于 y 求不定积分, 得

$\ln u = t(x) + I(y)$, 因此 $u(x,y) = f(x)g(y)$. 证毕

6. 设 $f(x,y) \in C^2(\mathbb{R}^2)$ 满足 Laplace 方程 $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$.

证明: $u(x,y) = f\left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$ 也满足 Laplace 方程.

证明: 因为 $\frac{\partial u}{\partial x} = f_1' \frac{y^2 - x^2}{(y^2 + x^2)^2} - f_2' \frac{2xy}{(y^2 + x^2)^2}$, $\frac{\partial u}{\partial y} = f_1' \frac{-2xy}{(y^2 + x^2)^2} + f_2' \frac{x^2 - y^2}{(y^2 + x^2)^2}$,

因此

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= f_{11}'' \frac{(y^2 - x^2)^2}{(y^2 + x^2)^4} - f_{12}'' \frac{4xy(y^2 - x^2)}{(y^2 + x^2)^4} + f_{22}'' \frac{4y^2 x^2}{(y^2 + x^2)^4} \\ &\quad + f_1' \frac{2x(y^2 + x^2)(x^2 - 3y^2)}{(y^2 + x^2)^4} + f_2' \frac{2y(y^2 + x^2)(3x^2 - y^2)}{(y^2 + x^2)^4}, \end{aligned}$$

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2} &= f''_{11} \frac{4y^2 x^2}{(y^2 + x^2)^4} - f''_{12} \frac{4xy(x^2 - y^2)}{(y^2 + x^2)^4} + f''_{22} \frac{(y^2 - x^2)^2}{(y^2 + x^2)^4} \\ &\quad + f'_1 \frac{2x(y^2 + x^2)(3y^2 - x^2)}{(y^2 + x^2)^4} + f'_2 \frac{2y(y^2 + x^2)(y^2 - 3x^2)}{(y^2 + x^2)^4},\end{aligned}$$

由于 $f''_{11} + f''_{22} = 0$, 故 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (f''_{11} + f''_{22}) \frac{(y^2 - x^2)^2}{(y^2 + x^2)^4} + (f''_{11} + f''_{22}) \frac{4y^2 x^2}{(y^2 + x^2)^4} = 0$. 证毕

7. 设 n 为整数, 若对任意的 $t > 0$, $f(tx, ty) = t^n f(x, y)$, 则称 f 是 n 次齐次函数。证明:

可微函数 $f(x, y)$ 是零次齐次函数的充要条件是 $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0$.

证明: 先证必要性。设可微函数 $f(x, y)$ 是零次齐次函数, 即

$$f(tx, ty) = f(x, y) \quad (\forall t > 0). \quad (4)$$

若 f 在坐标原点处有定义, 则由 f 的连续性可知 $f(x, y) = f(0, 0)$, $(\forall (x, y))$. 结论显然成立。

现在假设 f 在坐标原点处没有定义。则由复合函数的链式法则, 方程(4)两边分别对 t 求

导, 得 $x \frac{\partial f}{\partial x}(tx, ty) + y \frac{\partial f}{\partial y}(tx, ty) = 0$. 令 $t=1$, 即得

$$x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = 0.$$

必要性得证。

下证充分性。设 $f(x, y)$ 满足 $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0$. 任取 $(x, y) \in \mathbb{R}^2$, 令 $\vec{r} = (x, y)$, 且

$x = r \cos \theta$, $y = r \sin \theta$, $r > 0$. 则

$$\frac{\partial f}{\partial \vec{r}} = \frac{\partial f}{\partial x} \cdot \cos \theta + \frac{\partial f}{\partial y} \cdot \sin \theta = \frac{1}{r} (x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}) = 0.$$

上式说明 f 沿着方向 $\vec{r}$ 的方向导数等于零, 从而 f 沿着方向 $\vec{r}$ 的函数值不变。故在极坐标

系中 f 只是 $\theta = \arctan \frac{y}{x}$ 的函数, 这等价于只是 $\frac{y}{x}$ 的函数。可记 $f(x, y) = \phi(\frac{y}{x})$. 显然 ϕ 是

零次齐次函数。证毕

8. 设 $f(x, y)$ 在 $P_0(x_0, y_0)$ 可微。已知 $\vec{v} = \vec{i} - \vec{j}$, $\vec{u} = -\vec{i} + 2\vec{j}$, 且 $\left. \frac{\partial f}{\partial \vec{v}} \right|_{P_0} = 2$, $\left. \frac{\partial f}{\partial \vec{u}} \right|_{P_0} = 1$,

求 $f(x, y)$ 在 $P_0(x_0, y_0)$ 的全微分。

解: 因为 $\vec{v} = \vec{i} - \vec{j} = (1, -1)$, $\vec{u} = -\vec{i} + 2\vec{j} = (-1, 2)$, 且 $f(x, y)$ 在 $P_0(x_0, y_0)$ 可微, 因此

$$2 = \left. \frac{\partial f}{\partial \vec{v}} \right|_{P_0} = (f'_x(P_0), f'_y(P_0)) \cdot \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} (f'_x(P_0) - f'_y(P_0)),$$

$$1 = \left. \frac{\partial f}{\partial \vec{u}} \right|_{P_0} = (f'_x(P_0), f'_y(P_0)) \cdot \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) = \frac{1}{\sqrt{5}} (-f'_x(P_0) + 2f'_y(P_0)),$$

由此解出 $f'_x(P_0) = 4\sqrt{2} + \sqrt{5}$, $f'_y(P_0) = 2\sqrt{2} + \sqrt{5}$. 所以 $f(x, y)$ 在 $P_0(x_0, y_0)$ 的全微分

$$df(P_0) = (4\sqrt{2} + \sqrt{5})dx + (2\sqrt{2} + \sqrt{5})dy.$$

9. 设 $f(x, y)$ 为可微函数, $\vec{l}_1, \vec{l}_2$ 是 $\mathbb{R}^2$ 上的一组线性无关的向量。试证: $f(x, y)$ 在任一点 $P(x, y)$ 沿任意向量 $\vec{l}$ 的方向导数 $f'_l(P)$ 必定能用 $f'_{l_1}(P)$ 与 $f'_{l_2}(P)$ 线性表示。

证明: 令 $\vec{l}_1 = (\cos \alpha_1, \cos \beta_1)$, $\vec{l}_2 = (\cos \alpha_2, \cos \beta_2)$.

因为 $f(x, y)$ 可微, 故

$$\begin{cases} f'_{l_1}(P) = f'_x(P) \cos \alpha_1 + f'_y(P) \cos \beta_1 = d_1 \\ f'_{l_2}(P) = f'_x(P) \cos \alpha_2 + f'_y(P) \cos \beta_2 = d_2. \end{cases}$$

由于 $\vec{l}_1, \vec{l}_2$ 线性无关, 因此由上式解出 $\begin{pmatrix} f'_x(P) \\ f'_y(P) \end{pmatrix} = \begin{pmatrix} \cos \alpha_1 & \cos \beta_1 \\ \cos \alpha_2 & \cos \beta_2 \end{pmatrix}^{-1} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$.

于是, 对任意的向量 $\vec{l} = (\cos \alpha, \cos \beta)$,

$$\begin{aligned} f'_l(P) &= f'_x(P) \cos \alpha + f'_y(P) \cos \beta = (\cos \alpha, \cos \beta) \begin{pmatrix} f'_x(P) \\ f'_y(P) \end{pmatrix} \\ &= (\cos \alpha, \cos \beta) \begin{pmatrix} \cos \alpha_1 & \cos \beta_1 \\ \cos \alpha_2 & \cos \beta_2 \end{pmatrix}^{-1} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \\ &= (a, b) \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = ad_1 + bd_2, \end{aligned}$$

其中 $(a, b) = (\cos \alpha, \cos \beta) \begin{pmatrix} \cos \alpha_1 & \cos \beta_1 \\ \cos \alpha_2 & \cos \beta_2 \end{pmatrix}^{-1}$. 证毕

10. 设 $f(x, y) = x^2 - xy + y^2$, $P_0(1, 1)$. 试求 $\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0}$, 并问: 在怎样的方向 $\vec{l}$ 上, 方向导数

$\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0}$ 分别取得最大值、最小值和零。

解：因为 $f(x, y)$ 可微，且 $f'_x(P_0) = (2x - y)|_{(1,1)} = 1$, $f'_y(P_0) = (2y - x)|_{(1,1)} = 1$,

因此对任意的单位向量 $\vec{l} = (\cos \alpha, \cos \beta)$, $\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = \cos \alpha + \cos \beta$.

当 $\vec{l} = (1, 1)$ 是梯度方向时, $\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = \sqrt{2}$ 达到最大;

当 $\vec{l} = (-1, -1)$ 时, $\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = -\sqrt{2}$ 达到最小;

当 $\vec{l} = (1, -1)$ 或 $\vec{l} = (-1, 1)$ 时, 即 $\alpha = \frac{3\pi}{4}$ 或 $\frac{7\pi}{4}$ 时, $\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = 0$.

11. 设 a, b 是实数, 函数 $z = 2 + ax^2 + by^2$ 在点 $(3, 4)$ 处的方向导数中, 沿 $\vec{l} = -3\mathbf{i} - 4\mathbf{j}$ 的方向

导数最大, 最大值为 10, 求 a, b .

解：由于函数可微, 因此函数沿着梯度方向的方向导数达到最大, 又

$$\left. \frac{\partial z}{\partial x} \right|_{(3,4)} = 6a, \quad \left. \frac{\partial z}{\partial y} \right|_{(3,4)} = 8b,$$

且方向导数的最大值为 10, 故 $\frac{1}{5}(-3, -4) = (\frac{6a}{10}, \frac{8b}{10})$. 即 $\begin{cases} \frac{6a}{10} = -\frac{3}{5} \\ \frac{8b}{10} = -\frac{4}{5} \end{cases}$ 所以 $\begin{cases} a = -1 \\ b = -1 \end{cases}$.

12. 设 $f(x, y) \in C^2(\mathbb{R}^2)$ 满足 $f'_x(x, y) = f'_y(x, y)$, 且 $f(x, 0) > 0$.

试证明：对任意的 $(x, y) \in \mathbb{R}^2$, 有 $f(x, y) > 0$.

证明：令 $\vec{l} = (1, -1)$. 则对任意的 $(x, y) \in \mathbb{R}^2$, 因为 $f'_x(x, y) = f'_y(x, y)$, 所以 $\left. \frac{\partial f}{\partial \vec{l}} \right|_{(x,y)} = 0$,

即函数 $f(x, y)$ 在任意一点沿方向 $\vec{l} = (1, -1)$ 的方向导数为零, 故函数 $f(x, y)$ 在任意一点沿

方向 $\vec{l} = (1, -1)$ 是常数, 即在直线 $x + y = c$ 上 $f(x, y)$ 是常数。

对任意的点 $(x, y) \in \mathbb{R}^2$, 总存在直线 $L: x + y = c$ 使得 $(x, y) \in L$, 所以

$f(x, y) = f(c, 0) > 0$. 证毕

13. 证明: 若 $\mathbb{R}^2$ 上的可微函数 $f(x, y)$ 满足 $xf'_x + yf'_y = 0$, 则 $f(x, y)$ 恒为常数。

证明: 对 $\forall (x, y) \in \mathbb{R}^2$, 令 $\vec{r} = (x, y)$, 且令 $x = r \cos \theta$, $y = r \sin \theta$, $r > 0$, 则由条件可得

$$\frac{\partial f}{\partial \vec{r}}(x, y) = f'_x(x, y) \cos \theta + f'_y(x, y) \sin \theta = \frac{1}{r}(xf'_x(x, y) + yf'_y(x, y)) = 0,$$

即 $f(r \cos \theta, r \sin \theta)$ 与 r 无关. 任意固定 θ , 令 $r \rightarrow 0^+$, 再由 $f(x, y)$ 的连续性得

$$f(x, y) = \lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = f(0, 0).$$

证法二、对任意固定的 $(x, y) \in \mathbb{R}^2$, 定义一元函数 $\varphi(t) = f(tx, ty)$ ($t \geq 0$). 则

$$\varphi'(t) = xf'_1(tx, ty) + yf'_2(tx, ty).$$

于是当 $t > 0$ 时, 由条件可得 $\varphi'(t) = \frac{1}{t}[txf'_1(tx, ty) + tyf'_2(tx, ty)] = 0$, 所以 $\varphi(t)$ 为常数 ($t > 0$).

故 $\varphi(t) \equiv \varphi(1)$, 即 $f(tx, ty) = f(x, y)$. 由 $f(x, y)$ 的连续性, 令 $t \rightarrow 0^+$ 取极限, 得

$$f(x, y) = \lim_{t \rightarrow 0^+} f(tx, ty) = f(0, 0).$$

证毕

14. 设 $f(x, y)$ 在区域 $D \subset \mathbb{R}^2$ 上具有连续的偏导数, $L: \begin{cases} x = x(t) \\ y = y(t) \end{cases} (a \leq t \leq b)$ 是 D 中的

一段曲线, L 的端点为 A, B . 假设 $x, y \in C^1[a, b]$ 且 $x'(t)^2 + y'(t)^2 \neq 0$ ($\forall t \in [a, b]$).

证明: 若 $f(A) = f(B)$, 则存在点 $P_0(x_0, y_0) \in L$ 使得 $f'_l(P_0) = 0$, 其中 $\vec{l}$ 是曲线 L 在 P_0

的单位切向量。

证明: 令 $g(t) = f(x(t), y(t))$, $a \leq t \leq b$. 不妨设 A, B 分别对应着 $t = a, t = b$. 则由条件可

知 $g(t)$ 可导, 且 $g(a) = g(b)$. 由洛尔定理, 存在 $\mu \in (a, b)$ 使得 $g'(\mu) = 0$. 故

$$g'(\mu) = f'_x(x(\mu), y(\mu))x'(\mu) + f'_y(x(\mu), y(\mu))y'(\mu) = 0.$$

取 $x_0 = x(\mu)$, $y_0 = y(\mu)$. 则 $P_0(x_0, y_0) \in L$. 令 $\vec{l} = \frac{(x'(\mu), y'(\mu))}{\sqrt{x'(\mu)^2 + y'(\mu)^2}}$, 所以

$$\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = f'_x(x(\mu), y(\mu)) \frac{x'(\mu)}{\sqrt{x'(\mu)^2 + y'(\mu)^2}} + f'_y(x(\mu), y(\mu)) \frac{y'(\mu)}{\sqrt{x'(\mu)^2 + y'(\mu)^2}} = 0. \text{ 证毕}$$

15. 设 $f(x, y) = \varphi(xy)$, 而函数 φ 满足: 存在 $\delta > 0, \mu > \frac{1}{2}$ 和 $C > 0$ 使得

$$|\varphi(t)| \leq C|t|^\mu (\forall t \in (-\delta, \delta)). \text{ 证明: } f(x, y) \text{ 在 } (0, 0) \text{ 可微.}$$

证明: 当 $t \rightarrow 0$ 时, 有 $0 \leq \left| \frac{\varphi(t)}{\sqrt{t}} \right| \leq C |t|^{\mu-\frac{1}{2}} \rightarrow 0$, 故 $\lim_{t \rightarrow 0} \left| \frac{\varphi(t)}{\sqrt{2t}} \right| = 0$. 又

$$f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = 0, \quad f'_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = 0,$$

因此当 $(x,y) \rightarrow (0,0)$ 时,

$$\begin{aligned} 0 &\leq \left| \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} \right| \\ &= \left| \frac{\varphi(xy)}{\sqrt{x^2 + y^2}} \right| \leq \left| \frac{\varphi(xy)}{\sqrt{2}|xy|} \right| \rightarrow 0, \end{aligned}$$

所以 $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} = 0$, 故 $f(x,y)$ 在 $(0,0)$ 可微。证毕

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以下供学有余力的同学参考。

16. 若 $f'_x(x,y)$, $f'_y(x,y)$ 在 $P_0(x_0, y_0)$ 的邻域内存在, 且两个偏导函数均在 $P_0(x_0, y_0)$ 点可

微, 则 $f''_{xy}(P_0) = f''_{yx}(P_0)$.

证明: 任意固定充分小的 $h \neq 0$, 令 $\varphi(x) = f(x, y_0 + h) - f(x, y_0)$ 且

$$g(h) = f(x_0 + h, y_0 + h) - f(x_0 + h, y_0) - f(x_0, y_0 + h) + f(x_0, y_0).$$

因为 $f'_x(x,y)$ 在 $P_0(x_0, y_0)$ 的邻域内存在, 故 $\varphi(x)$ 可导且

$$\varphi'(x) = f'_x(x, y_0 + h) - f'_x(x, y_0).$$

由一元函数的微分中值定理, 存在 $0 < \theta_1 < 1$ 使得

$$\begin{aligned} g(h) &= \varphi(x_0 + h) - \varphi(x_0) = \varphi'(x_0 + \theta_1 h)h \\ &= (f'_x(x_0 + \theta_1 h, y_0 + h) - f'_x(x_0 + \theta_1 h, y_0))h. \end{aligned}$$

又因为 $f'_x(x,y)$ 在 $P_0(x_0, y_0)$ 可微, 故当 $h \rightarrow 0$ 时,

$$f'_x(x_0 + \theta_1 h, y_0 + h) = f'_x(x_0, y_0) + f''_{xx}(x_0, y_0)\theta_1 h + f''_{xy}(x_0, y_0)h + o(h),$$

且 $f'_x(x_0 + \theta_1 h, y_0) = f'_x(x_0, y_0) + f''_{xx}(x_0, y_0)\theta_1 h + o(h)$, 从而

$$g(h) = f''_{xy}(x_0, y_0)h^2 + o(h^2) \quad (5)$$

同理, 讨论可导函数 $\psi(y) = f(x_0 + h, y) - f(x_0, y)$, 存在 $0 < \theta_2 < 1$ 使得

$$\begin{aligned} g(h) &= \psi(y_0 + h) - \psi(y_0) = \psi'(y_0 + \theta_2 h)h \\ &= (f'_y(x_0 + h, y_0 + \theta_2 h) - f'_y(x_0, y_0 + \theta_2 h))h. \end{aligned}$$

由 $f'_y(x, y)$ 在 $P_0(x_0, y_0)$ 可微得到

$$g(h) = f''_{yx}(x_0, y_0)h^2 + o(h^2) \quad (h \rightarrow 0) \quad (6)$$

由(5)式与(6)式, 得 $f''_{xy}(P_0) = \lim_{h \rightarrow 0} \frac{g(h)}{h^2} = f''_{yx}(P_0)$. 证毕

17. 设 $f'_x(x, y)$, $f'_y(x, y)$ 在 $P_0(x_0, y_0)$ 的邻域内存在, 且 f''_{xy} 在 $P_0(x_0, y_0)$ 连续, 则 $f''_{yx}(P_0)$ 存

在且 $f''_{xy}(P_0) = f''_{yx}(P_0)$.

证明: 对任意固定的充分小的 $\Delta y \neq 0$, 令

$$\varphi(x) = f(x, y_0 + \Delta y) - f(x, y_0).$$

则 $\varphi(x)$ 可导, 且

$$\begin{aligned} \Delta &= \varphi(x_0 + \Delta x) - \varphi(x_0) \\ &= f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) - f(x_0, y_0 + \Delta y) + f(x_0, y_0). \end{aligned}$$

由一元函数的微分中值定理, 存在 $0 < \theta_1 < 1$ 使得

$$\Delta = \varphi'(x_0 + \theta_1 \Delta x) = (f'_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) - f'_x(x_0 + \theta_1 \Delta x, y_0))\Delta x.$$

再一次应用一元函数的微分中值定理, 存在 $0 < \theta_2 < 1$ 使得

$$\Delta = f''_{xy}(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y)\Delta x \Delta y.$$

这样

$$\begin{aligned} &\frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0)}{\Delta y} - \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} \\ &= f''_{xy}(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y)\Delta x, \end{aligned}$$

从而两边对 $\Delta y \rightarrow 0$ 时取极限, 得

$$f'_y(x_0 + \Delta x, y_0) - f'_y(x_0, y_0) = f''_{xy}(x_0 + \theta_1 \Delta x, y_0)\Delta x.$$

$$\text{故 } \lim_{\Delta x \rightarrow 0} \frac{f'_y(x_0 + \Delta x, y_0) - f'_y(x_0, y_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} f''_{xy}(x_0 + \theta_1 \Delta x, y_0) = f''_{xy}(x_0, y_0).$$

所以 $f''_{yx}(x_0, y_0)$ 存在且 $f''_{yx}(x_0, y_0) = f''_{xy}(x_0, y_0)$. 证毕

18. 设定义在开区域 $D \subset \mathbb{R}^2$ 上的可微函数 $f(x, y)$ 满足 $\nabla f(x, y) = 0$ ($\forall (x, y) \in \mathbb{R}^2$), 证

明: $f(x, y)$ 是 D 上的常值函数。

证明: 若区域 D 是凸集, 则由本习题 13 知, $f(x, y)$ 恒为常数。对于一般的开区域 D , 在 D

中任取两点 $\mathbf{x}_0$ 和 $\mathbf{x}$, 下面证明 $f(\mathbf{x}) = f(\mathbf{x}_0)$.

因为 D 是连通集, 因此存在连结两点 $\mathbf{x}_0$ 和 $\mathbf{x}$ 的连续曲线 $\gamma: [0, 1] \rightarrow D$ 使得 $\gamma(0) = \mathbf{x}_0$, $\gamma(1) = \mathbf{x}$. 又知 D 是开集, 因此存在 $\delta > 0$ 使得 $\mathcal{B}(\mathbf{x}_0, \delta) \subset D$. 因为 γ 连续, 所以存在 $\eta > 0$ 使得 $\gamma([0, \eta]) \subset \mathcal{B}(\mathbf{x}_0, \delta)$. 显然 $\mathcal{B}(\mathbf{x}_0, \delta)$ 是凸集, 故 $f(x, y)$ 在 $\mathcal{B}(\mathbf{x}_0, \delta)$ 上是常数. 所以 $f(\gamma(\eta)) = f(\mathbf{x}_0)$. 令

$$t_0 = \sup\{t \mid t \in [0, 1], f(\gamma(t)) = f(\mathbf{x}_0)\}.$$

假设 $t_0 < 1$. 因为 $\gamma(t_0) \in D$, 因此存在 $\delta' > 0$ 使得 $\mathcal{B}(\gamma(t_0), \delta') \subset D$. 由上面的讨论知, 存在 $\eta' > 0$ 且 $t_0 + \eta' < 1$ 使得 $\gamma(t_0 + \eta') \in \mathcal{B}(\gamma(t_0), \delta')$, 所以 $f(\gamma(t_0 + \eta')) = f(\gamma(t_0)) = f(\mathbf{x}_0)$, 这与 t_0 的定义矛盾, 所以 $t_0 = 1$ 且 $f(\mathbf{x}) = f(\gamma(1)) = f(\mathbf{x}_0)$. 证毕